

Adaptive AI

Note: This article is intended as a reference, contributing to the strong development of the AI community in Vietnam. I hope that those who read this piece will be inspired to innovate in the field and bring real value to businesses and society at large.

Greetings, readers. My name is Nguyen Duc Tri, currently a third-year Computer Science student at Ha Noi University of Science and technology, under the Troy IT joint program of the United States. I come from a family without a background in IT, but we are a cultured family, educated in a supportive environment, and have achieved modest accomplishments. Though my major is Computer Science, I have a deep passion for the cybersecurity – a field that, while not highly prominent in recent years, plays a pivotal role across all sectors and is one of the most important industries of the future.

Today, cybersecurity is undergoing powerful transformations—both positive and negative. As cyberattack methods grow increasingly cunning and complex, cybersecurity faces real challenges. Since the launch of ChatGPT in 2022, followed by the explosive growth of AI technologies, applying AI to work has become inevitable—even for cybercriminals. That is why I dared to look directly at this reality and began developing new visions to protect users online and, at a higher level, to safeguard enterprises.

While we may not immediately eradicate cybercrime, we can design effective models that meet the essential needs of citizens and the economic activities of businesses. Adaptive AI is a personal project I began in late April 2025 and is currently under development. For me, this is a long-term project with potential impact on the cybersecurity community. My aim is non-profit, with hopes to deploy it in the digital infrastructure of small and medium-sized enterprises in Vietnam. This may be an ambitious goal for a young and relatively inexperienced student like me, but I hope passionate developers will find opportunities to grow in this field, contributing to safety and sustainability in the digital era.

Part I

What is Adaptive AI? Definitions and Explanations

Adaptive AI is a project focused on developing AI technologies toward original automation—an automatic defense system from deep within the operating system, reinforcing aging

structures that lack sufficient cybersecurity infrastructure. The word "adaptive" refers to flexibility: the system can respond quickly to real-world scenarios in real time. The key idea is to *not* use traditional modern approaches like NLP, LLMs, or ML.

Why take this route? Let's examine some basic characteristics of large language models (LLMs) and popular machine learning algorithms:

Large Language Models:

1. **Probabilistic Modeling:** Words are predicted based on conditional probabilities $P(w_t | w_{<t})$, allowing fluid generation of text but making accuracy difficult to control.
2. **Transformer Architecture:** Efficient for long-range context handling, outperforming RNNs/LSTMs.
3. **Pre-training and Fine-tuning:** Learning from masked/casual language modeling, followed by task-specific tuning.
4. **Embeddings:** Words and sentences are mapped into high-dimensional dense vectors that encode logic, causality, and social relations.
5. **Generalization:** Due to scale and diversity of data, LLMs can answer unseen questions.
6. **Limited Adaptivity:** LLMs follow preset models and struggle with out-of-distribution data.
7. **Resource Cost:** LLMs require vast resources and create energy and carbon burdens.

Machine Learning:

1. **Linear Regression:** Simple and interpretable, but unsuitable for non-linear data.
2. **Logistic Regression:** For binary classification using sigmoid outputs.
3. **SVMs:** Optimal separating hyperplanes; kernel trick for non-linear problems.
4. **Decision Trees and Random Forests:** Interpretable but prone to overfitting; ensembles help.
5. **Gradient Boosting:** (XGBoost, LightGBM) Very effective for tabular data.

Key Weaknesses:

LLMs and ML models are computation-heavy, requiring costly infrastructure. Fast AI often

sacrifices user privacy, which can be exploited for phishing, MITM attacks, etc. Adaptive AI mitigates these risks while maintaining reliability and trust.

Core Philosophy of Adaptive AI (10 Statements):

1. Adaptive AI changes based on real-world context.
 2. It learns only what is needed, at the right time/place.
 3. It doesn't just react—it anticipates and adapts.
 4. Change doesn't mean chaos; it implies guided stability.
 5. It is not just a model, but a living logical entity.
 6. It exists between automation and intuition.
 7. "Adaptivity" isn't randomness.
 8. It learns from both internal (data) and external (context) sources.
 9. Adaptive AI is software that lives.
 10. True adaptation has ethics—without it, it becomes a monster.
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Part II

Technical Design of Adaptive AI

Core Features:

- Modular and replaceable components, with internal evaluation mechanisms.
- Integration of meta-learning to update both old and new logic.
- Conscious state modeling (State), applied with an Engine to analyze real-time context.
- Behavior engine with anti-manipulation logic (Trust-behavior).
- Logic learning via plugin system:
Input logic [1] → compare old/new[2] → generate behavior[3] → execute[4] → verify[5]
→ defend[6] → output.
- Kill-switch for unethical logic and rollback mechanisms.

Modeling Techniques:

- Bayesian Optimization (future plan).
- Thompson Sampling for behavioral choices.

Data and Knowledge:

- Can handle unlabeled data.
- Distinguishes noise vs. signal.
- Favors relevant over big data.
- Information layered: raw → learning → action → evaluation.
- Temporary memory for instant reaction.
- Knowledge = experience + context + timeline.

Security and Survival:

- Zero trust behavior.
- Can self-lock when misused.
- Detects forced unethical commands.
- Immune to prompt injection.
- Alerts on risk thresholds.
- Operates only within assigned privileges.

Modularity and Integration:

- Modules replaceable without affecting whole (SOLID).
- Can support distributed servers.
- Uses low-level (C++23, C) and procedural high-level (Lua).
- API interactions (Windows, WebSocket).
- Wireshark packet experiments.
- Cross-platform build (Windows, Linux).
- Debug-friendly and updatable without full rewrites.
- Kernel embeddable.

Software Architecture:

- Clear core APIs (*.h).
- Behavioral logging.
- Compile-time support.
- CPU, memory, time usage policies.

Vision:

- Human-interactive GUI.
 - Emergency backup servers.
 - Real-time monitoring.
 - No sandbox dependency.
 - Rejects junk data.
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Part III

Scope of Training

"If a being can only learn what it's allowed to, can it ever surpass its teacher?"

Training scope is not just about data but permission to learn and transcend limits.

AI or humans can't grow if:

- Locked in rigid frameworks.
- Isolated from paradox.
- Forbidden to err and relearn.

In Adaptive AI, training scope includes behavioral rules, states, reactions. At the abstract level: generalizing from unseen states. It estimates hidden probabilities and acts invisibly. Meta-learning score: learning methods not taught, inferred from past tasks and logic models. Unlike ML which is limited to code/datasets, Adaptive AI designs open systems that birth new knowledge.

Conclusion

Unlike patching from the outside-in, the system is:

- Secure from the kernel level.
- Cloud/vendor independent.
- Self-monitoring in real-time, resilient against failure.